

Selecting Optimal Unrestricted Hartree–Fock Trial Wave Functions for Phaseless Auxiliary-Field Quantum Monte Carlo: Accuracy and Limitations in Modeling Three Iron–Sulfur Clusters

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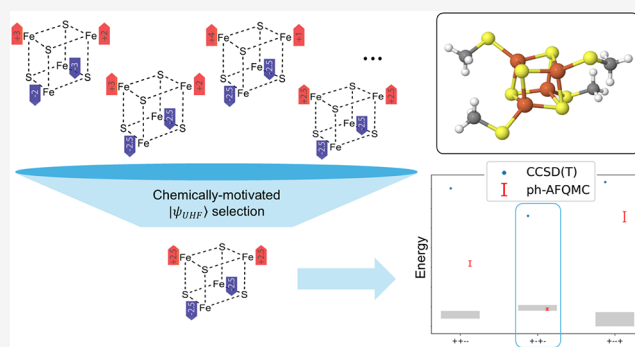


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ABSTRACT: Phaseless auxiliary-field quantum Monte Carlo (ph-AFQMC) has emerged as a promising electronic structure method for correlated electronic systems. However, the quality of its predictions depends critically on the choice of trial wave function, and it is not obvious how to make an optimal choice, especially for strongly correlated states of large systems. Mean-field wave functions are compelling trial wave function candidates, as they map directly to chemical concepts and can be obtained with $O(N^4)$ cost. Yet in the strongly correlated regime, one faces a symmetry dilemma and the existence of multiple nearly degenerate solutions. In this work, we investigate active space models of $[2\text{Fe}-2\text{S}]^{2+}$, mixed-valent $[4\text{Fe}-4\text{S}]^{2+}$, and $[4\text{Fe}-4\text{S}]^{4+}$ and explore the sensitivity of ph-AFQMC to the choice of unrestricted Hartree–Fock trial wave function. We find that surprisingly accurate ground-state energies for these systems can be obtained when trial selection is guided by chemical properties and physical symmetries, rather than the variational energy. However, in all cases, we find a rapidly decaying overlap between the stochastic wave function and the UHF trial, indicating that the trials are acting as suboptimal importance functions. By analogy to a similar situation in the stretched helium dimer cation, we show how this sampling bias pushes ph-AFQMC with UHF trials toward artificially negative energies, which evidently can be compensated for by the phaseless bias in certain cases.



1. INTRODUCTION

It has been known since the 1950s¹ that polynuclear transition metal clusters have significant biological relevance across a wide range of redox reactions.^{2–5} Prominent examples of their electrocatalytic roles include oxygen evolution at the tetramanganese oxygen evolving complex (OEC) in Photosystem II,^{6,7} electron transfer at di- and tetra-nuclear iron–sulfur clusters,⁸ and nitrogen reduction via the FeMo cofactor in nitrogenase enzymes.⁹ Alongside advances in experimental techniques, tremendous efforts have been made using computational quantum chemistry methods such as density functional theory (DFT) and related broken-symmetry approaches to elucidate magnetic and structural properties along with reaction mechanisms.^{10–18}

However, it is also known that polynuclear transition metal clusters pose a challenge for approximate electronic structure models based on the mean-field formalism, including DFT. The presence of multiple entangled spins can give rise to strong electron correlation¹⁹ in catalytically relevant low-energy states, which requires multiconfigurational wave functions²⁰ (though many can be qualitatively captured by one or a relatively small number of Configuration State Functions^{21–25}). In this regime, a growing number of studies have found that DFT is very

sensitive to the choice of exchange–correlation functional,^{26–28} which motivates the need to develop and employ more reliable *ab initio* many-body techniques.

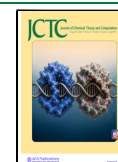
Chan and coworkers performed the first many-body wave function studies of the OEC²⁹ and di- and tetra-nuclear Fe–S clusters³⁰ using the Density Matrix Renormalization Group (DMRG). More recently, DMRG studies of other iron–sulfur clusters in nitrogenase, including the P-cluster³¹ and FeMoco,³² have been reported. These studies provide *ab initio* descriptions of the many-body wave functions and properties such as spin–spin correlations and natural orbital occupation numbers, albeit in reduced active space models containing only the Fe 3d orbitals and S 3p orbitals. Systematic extrapolation of the bond-dimension parameter enabled near-exact ground-state energy estimates. Among their major conclusions is that an effective

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interacting spin Hamiltonian does not always capture the low-energy physics, which has important implications regarding the validity of widely used strategies such as broken-symmetry DFT that are based on Ising/Heisenberg spin physics. These Fe–S active space models have become popular targets for quantum computation/simulation due to their complexity and relatively small system sizes, especially in the 25–100 logical qubit regime.^{31,33–36} There is also an impressive body of work by Li Manni and coworkers, which pioneers the use of spin-adapted multiconfigurational wave functions, realized via compression techniques^{22,37} or a stochastic FCIQMC approach,³⁸ to polynuclear transition metal clusters such as the oxygen-evolving complex³⁹ and iron–sulfur clusters.^{38,40} Multireference wave function methods have also been combined with more approximate treatment of the protein environment in, e.g., a recent study by Pantazis and coworkers.⁴¹

Another emerging electronic structure method, which has the ability to scale up to hundreds and even thousands of orbitals with typical trial wave functions is phaseless auxiliary-field quantum Monte Carlo (ph-AFQMC).^{42–45} Initially developed in the condensed matter physics community in the context of lattice models, ph-AFQMC has undergone intense development in the quantum chemistry community in recent years. While some consensus is emerging regarding the requisite polynomial-cost trial wave functions for the accurate description of weakly correlated molecular systems – including restricted Hartree–Fock,^{45,46} coupled cluster wave functions projected onto CI subspaces,^{47,48} and strategies involving single-shot selected CI calculations following active-space orbital optimization⁴⁹ – there is still a lack of consensus for strongly correlated electronic states in, e.g., polynuclear transition metal compounds. The myriad approaches of multiple active research groups to obtaining the trial might be classified into two classes. The first relies on multideterminant trial wave functions that approximate the exact wave function, including truncated full CI,⁵⁰ selected CI trials,^{51,52} and matrix product states.⁵³ The second posits that spin symmetry-breaking at the mean-field level can provide a sufficiently appropriate trial wave function for certain classes of strongly correlated states.^{54,55} Indeed, from a computational cost perspective the latter approach is clearly preferable as the cost, in general, of the former trial wave functions scales exponentially with system size. However, as will be explored in this work, a critical challenge of the second approach is the fact that Hartree–Fock (even in the simplest molecular models of strong correlation) has a complex solution landscape littered with myriad nearly degenerate local minima.⁵⁶ ph-AFQMC can be highly sensitive to the choice of solution used as the trial wave function; for example, in a recent work we examined the dissociation of O₂ with ph-AFQMC-GHF⁵⁴ and identified multiple GHF solutions separated by *O*(1) mHa that led to an unexpectedly wide range of ph-AFQMC energies. The situation is much more severe in polynuclear transition metal compounds, *vide infra*.

The motivating question that the present work endeavors to address is of the combinatorially many mean-field solutions that exist in strongly correlated states, how can one choose the most optimal unrestricted Hartree–Fock wave function as a trial for ph-AFQMC or as a reference state for low-order coupled cluster models? Our data suggests that the answer is *not* necessarily the single determinant with the lowest variational energy, although many times this is the case. Corroborating prior work focused on molecular nonvalence correlation-bound anions,⁵⁷ we find that quantities reflecting the chemically or physically sensible

distribution of electron density are more useful descriptors – in this work we use atomic spin densities (or the dipole moment for the linear He₂⁺ model) to pinpoint the trial or reference state that best reflects either rigorous physical symmetries or oxidation state assignments/spin coupling patterns supported by experimental data. In both ph-AFQMC and UCCSD(T) this leads to the most accurate ground-state energies vs either exact FCI energies or theoretical best estimates.

However, the nonmonotonic energy trajectory that eventually lands on an overly negative ph-AFQMC energy observed for the [4Fe–4S]⁴⁺ state motivated us to investigate the evolution of walkers under the phaseless constraint. Our analysis indicates that, in analogy with the stretched He₂⁺ case, the stochastic wave functions in all three Fe–S systems investigated exhibit an overlap with the trial wave function that quickly decays toward zero, which suggests suboptimal guidance of the walker trajectories in imaginary time by the trial state (importance sampling) and that local energies are, on average, spuriously negative. This explains the, at times, overly negative ph-AFQMC energies, and the contemporaneous finding of Chan and coworkers that ph-AFQMC energies for Fe–S systems can become more positive when correlated trial wave functions are used.⁵⁸

2. METHODS

2.1. The Phaseless Constraint

Imaginary-time propagation of an initial state can be exactly mapped onto an open-ended random walk in nonorthogonal Slater determinant space.^{42,43,45,59} The Monte Carlo wave function $|\Psi_{\text{MC}}(\tau)\rangle$, defined at a point in imaginary-time τ , can be expressed as a linear combination of random walkers, each (indexed i) with a weight $w_i(\tau)$, a determinant $|\varphi_i(\tau)\rangle$ and a trial wave function $|\varphi_T\rangle$:

$$|\Psi_{\text{MC}}(\tau)\rangle \approx \sum_i w_i(\tau) \frac{|\varphi_i(\tau)\rangle}{\langle \varphi_T | \varphi_i(\tau) \rangle} \quad (1)$$

The imaginary-time propagator integrates over the product of an effective one-body propagator $\hat{B}(\mathbf{x})$ and a normal distribution $N(\mathbf{x})$ over auxiliary fields $\mathbf{x}$:

$$\exp(-\Delta\tau\hat{H}) \approx \int_{-\infty}^{\infty} N(\mathbf{x}) \hat{B}(\mathbf{x}) d\mathbf{x} \quad (2)$$

This equation can be modified with an importance sampling transformation, where the integration variable is shifted by a so-called force bias, $\bar{\mathbf{x}}$, chosen such that the phase of the overlap ratio, $\frac{\langle \varphi_T | \hat{B}(\mathbf{x} - \bar{\mathbf{x}}) | \varphi_i \rangle}{\langle \varphi_T | \varphi_i \rangle}$, vanishes at *O*($\sqrt{\Delta\tau}$). The weight update factor is

$$\frac{\langle \varphi_T | \varphi_i(\tau) \rangle}{\langle \varphi_T | \varphi_i(\tau - \Delta\tau) \rangle} * e^{\frac{1}{2}\bar{\mathbf{x}}^2 - \mathbf{x}\bar{\mathbf{x}}} \approx e^{-\Delta\tau E_L} \quad (3)$$

where the local energy is $E_L(\tau) = \frac{\langle \varphi_T | \hat{H} | \varphi_i(\tau) \rangle}{\langle \varphi_T | \varphi_i(\tau) \rangle}$. The “hybrid” formalism uses the left-hand side of eq 3 explicitly, and the specific choice of the force-bias ensures that the phases of these two factors cancel to leading order.⁶⁰

However, even when using small imaginary-time steps and the weight update in eq 3, MC estimates will still inevitably suffer from an exponentially vanishing signal-to-noise ratio. To understand this, consider the MC estimate of the energy:

$$E_{\text{MC}}(\tau) = \frac{\sum_i w_i(\tau) E_L^i(\tau)}{\sum_i w_i(\tau)} \quad (4)$$

Each walker weight is a stochastically fluctuating complex number that acquires an arbitrary phase. For systems where the propagator and thus walker weights and orbitals are real-valued, the phase of each walker

weight can be only 0 or π , leading to weights being either along the positive or negative real axis. When E_{MC} is calculated the contributions from walkers with positive and negative weights will tend to cancel, resulting in a loss of MC signal. The constrained-path approximation keeps only walkers with a positive overlap with a fixed trial wave function.⁵⁹ In the case of phaseless AFQMC, the phaseless constraint involves taking the norm of the weight update factor (or real part of the local energy) in eq 3 and then multiplying this by $\max(0, \cos(\Delta\theta))$,

where $\Delta\theta = \arg\left[\frac{\langle\psi_T|\psi_i(\tau)\rangle}{\langle\psi_T|\psi_i(\tau-\Delta\tau)\rangle}\right]$. The phaseless constraint, which relies

on optimal choice of the force bias to keep (unconstrained) phase rotations of the weight update factor small, keeps track of the overlap of the walker determinants with a fixed trial wave function to break the rotational invariance in the complex plane (i.e., φ and $e^{i\gamma}\varphi$ are physically identical). It defines a gauge choice ($\gamma = 0$) such that the $\{w_i\}$ in eq 1 are constrained to be real and positive, and walker determinants that accumulate a phase in excess of $\pm\pi/2$ are removed. In the zero time step limit, the phaseless formalism was designed to prevent a finite population at the origin of the complex plane defined by $\langle\psi_T|\varphi\rangle$, but in practice this cannot be strictly enforced.

When the trial wave function is an exact eigenstate of the Hamiltonian, the phaseless constraint does not bias the ph-AFQMC energies. For weakly correlated systems, as the trial approaches the limit of exactness the accuracy of ph-AFQMC has been shown to improve systematically; however, in the regime of strong correlation this may not always be the case. In this work we explore the class of unrestricted Hartree–Fock (UHF) trial states. For the Fe–S clusters investigated we find that a judicious choice of UHF trial enables ph-AFQMC energies to be very close to the reference values, despite a rapid decay in the overlap between the stochastic MC wave function and the trial.

2.2. Fe–S Models

Extensive experimental efforts have been devoted to the synthesis and characterization of Fe–S cluster models^{8,61} that mimic the chemical and physical motifs most commonly found in biological systems.² Early demonstrations of synthetic Fe–S clusters in the 1970s⁶² have been followed by a wide array of different model systems.⁶³ Multiple spectroscopic techniques including electron paramagnetic resonance (EPR),⁶⁴ Mossbauer,⁶⁵ and nuclear magnetic resonance (NMR)⁶⁶ have been essential for deducing the structure and properties of these clusters since the earliest investigations of Fe–S proteins. Beyond experiment, phenomenological models of the $[2\text{Fe}–2\text{S}]^{67–70}$ and $[4\text{Fe}–4\text{S}]^{71,72}$ clusters have likewise been used to explore the electronic and magnetic character of these states. The history and limitations of these models have been well reviewed in a more recent *ab initio* investigation of the complexes,³⁰ a work which corroborates the prediction that the iron centers of the $[2\text{Fe}–2\text{S}]^{2+}$ cluster are antiferromagnetically coupled.^{73–76}

For the $[4\text{Fe}–4\text{S}]^{2+}$ cluster, while simple electron counting would suggest the presence of two Fe(III) ions and two Fe(II) ions, spectroscopic experiments find that each Fe oxidation state is effectively 2.5.⁷⁷ Structural studies have established that the cubane is distorted along one axis, and the pair of Fe atoms on the top or bottom surface of the cube is ferromagnetically coupled into an $S = \frac{9}{2}$ state, as indicated in Figure 2. The intrapair ferromagnetic coupling has been explained by a “double exchange” mechanism,^{78–82} and the two formally mixed-valence irons in a ferromagnetic pair are experimentally found to be indistinguishable, implying even charge delocalization between the two-centers. In turn, the two ferromagnetic pairs (one on the top face, the other on the bottom face) are antiferromagnetically coupled resulting in an overall spin singlet ground-state.^{83,80,70}

The third system we study is the highly oxidized $[4\text{Fe}–4\text{S}]^{4+}$ state, with four antiferromagnetically coupled Fe(III) centers. While irrelevant to biological Fe–S processes, it has been accessed experimentally in a synthetic cluster⁸³ and its properties have been explored theoretically.^{36,84}

We have used the 30e20o $[2\text{Fe}–2\text{S}]^{2+}$ and 54e36o $[4\text{Fe}–4\text{S}]^{2+}$ active space models (based on localized high-spin DFT orbitals) as provided by Li and Chan.⁸⁵ We used the DMRG reference for $[2\text{Fe}–$

$2\text{S}]^{2+}$ from the same report, however for $[4\text{Fe}–4\text{S}]^{2+}$ a more recent work by Zhai et al.³² has suggested theoretical best estimates for three low-lying spin isomers based on extrapolated unrestricted DMRG and relatively high-order coupled cluster calculations. For the $[4\text{Fe}–4\text{S}]^{4+}$ state we used the 52e36o model and converged DMRG reference energy from Lee et al.³⁶ For this last active space system we unfortunately do not have geometric or orbital information, as we simply performed calculations starting from an FCIDUMP file. We determined the 20 Fe 3d orbitals used in the spin density analysis from the diagonal of the one-body Hamiltonian.

2.3. Computational Details

To locate the various stable UHF solutions (that were later used as references in coupled cluster and trials in ph-AFQMC) we used PySCF’s⁸⁶ Newton solver and stability analysis. We initialized our UHF solver with a one-electron reduced density matrix guess based on one of the following two strategies:

- Manual specification of the orbital occupancies (generated by enumerating all the different possible combinations of electrons on the iron atoms)
- Randomized guess generated by performing 10 random Givens rotations (each with an equal probability to occur either in the α or β sector of the density) between random orbitals with an angle chosen by drawing from a normal distribution $N(\pi/2, 1)$

Note that in our solution searches we discarded isoenergetic solutions (within 1 μHa) and limited correlated calculations to a low-energy subset of solutions.

To further “fingerprint” the UHF solutions we consider the net spin, magnitude, and orientation of the spin density operator:

$$\hat{\mathbf{s}}(\mathbf{r}) = \frac{1}{2} \sum_{\sigma\sigma'} \hat{\psi}_{\sigma}^{\dagger}(\mathbf{r}) \boldsymbol{\sigma}_{\sigma\sigma'} \hat{\psi}_{\sigma'}(\mathbf{r}) \quad (5)$$

The components of the expectation value of this vector may be obtained by contraction of the density

$$\rho(\mathbf{r}) = \begin{pmatrix} \langle\hat{\psi}_{\alpha}^{\dagger}(\mathbf{r})\hat{\psi}_{\alpha}(\mathbf{r})\rangle & \langle\hat{\psi}_{\beta}^{\dagger}(\mathbf{r})\hat{\psi}_{\alpha}(\mathbf{r})\rangle \\ \langle\hat{\psi}_{\alpha}^{\dagger}(\mathbf{r})\hat{\psi}_{\beta}(\mathbf{r})\rangle & \langle\hat{\psi}_{\beta}^{\dagger}(\mathbf{r})\hat{\psi}_{\beta}(\mathbf{r})\rangle \end{pmatrix} = \begin{pmatrix} \mathbf{P}_{\alpha\alpha} & \mathbf{P}_{\alpha\beta} \\ \mathbf{P}_{\beta\alpha} & \mathbf{P}_{\beta\beta} \end{pmatrix}$$

with the components of the Pauli matrices $\boldsymbol{\sigma}_{\sigma\sigma'}$:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (6)$$

The symmetry restrictions in ROHF and UHF cases ensure that the off-diagonal elements are $\mathbf{P}_{\alpha\beta} = \mathbf{P}_{\beta\alpha}^{\dagger} = 0$, and formally only the S_z contribution of the spin density vector has a surviving term through contraction with σ_z :

$$\hat{\mathbf{s}}(\mathbf{r})_z = \text{Tr} \left[\begin{pmatrix} \mathbf{P}_{\alpha\alpha} & 0 \\ 0 & \mathbf{P}_{\beta\beta} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right] = \mathbf{P}_{\alpha\alpha} - \mathbf{P}_{\beta\beta} \quad (7)$$

As the S_x and S_y components are exactly zero for these cases, this immediately determines the total net spin and the unidirectional orientation/magnitude of the collinear case for each UHF trial determinant under study.

The number of electrons of a specific spin may be obtained by the partial trace over the orbitals μ belonging to atomic center A , such that the Mulliken spin density of the center is determined by

$$\rho_s(A) = \sum_{\mu \in A} (\mathbf{P}_{\alpha\alpha} - \mathbf{P}_{\beta\beta})_{\mu\mu} S_{\mu\mu} \quad (8)$$

where we have contracted the surviving elements of the spin density vector $\hat{\mathbf{s}}(\mathbf{r})_z$ with the (often orthonormal) elements of the overlap matrix $\mathbf{S}$ associated with A . The Mulliken spin density has historically been related to the gross electron population and the free valency on atomic centers^{87,88} and has seen demonstrable success in characterizing oxidation states.^{89,90}

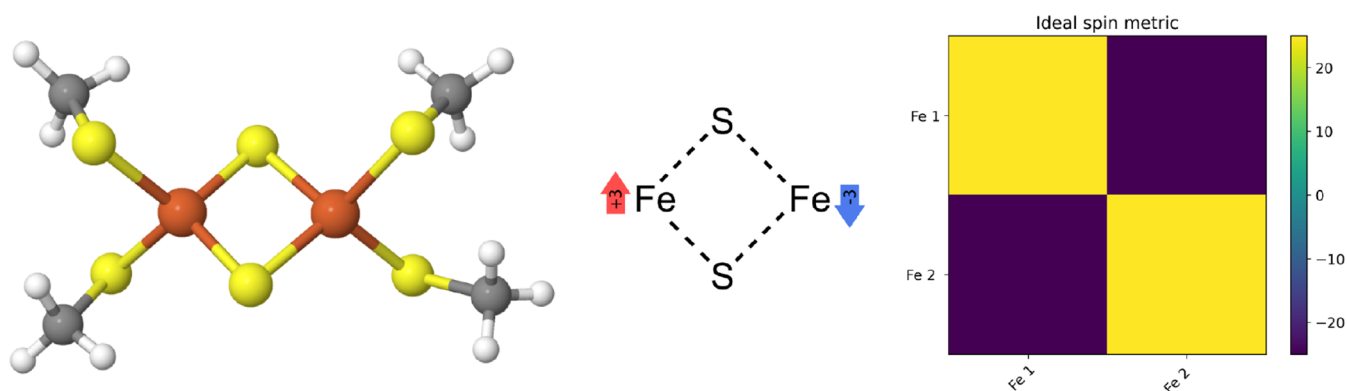


Figure 1. Cluster geometry (left), idealized spin configuration (middle), and corresponding reference spin density metric (right) for $[2\text{Fe}-2\text{S}]^{2+}$.

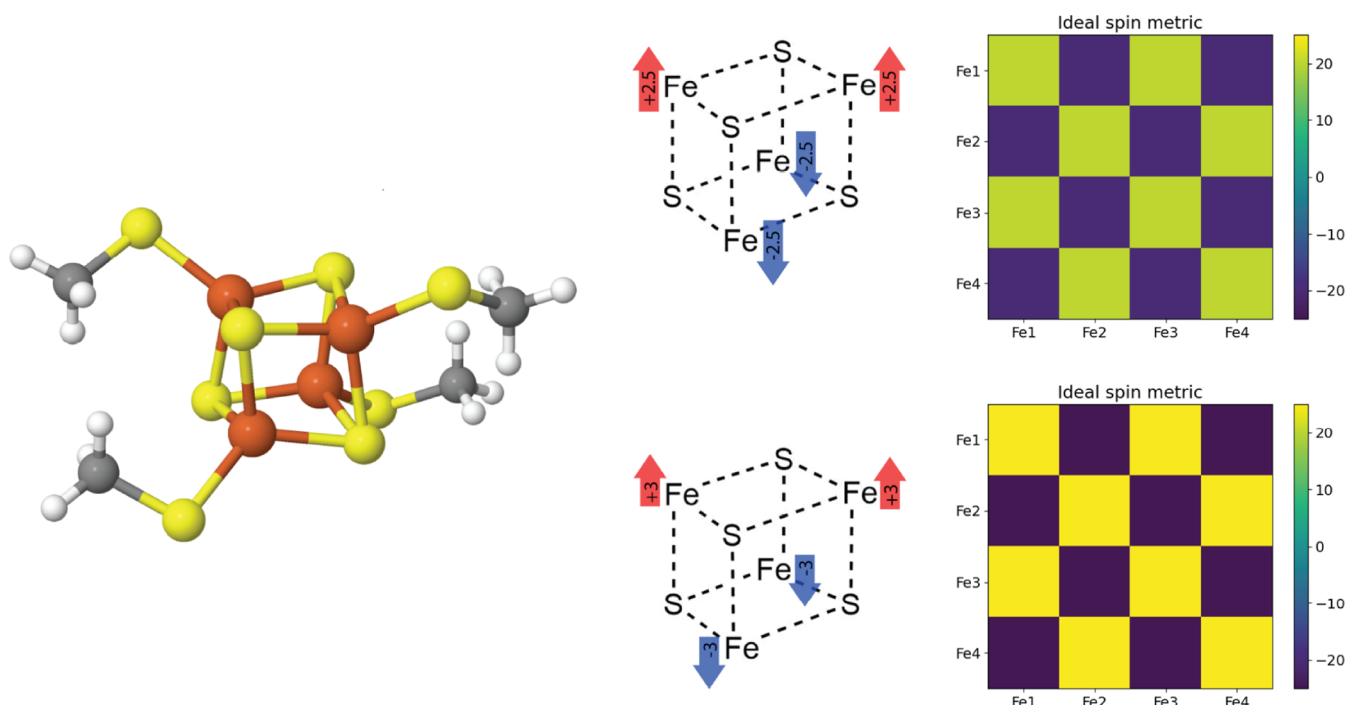


Figure 2. Same as Figure 1 but for $[4\text{Fe}-4\text{S}]^{2+}$ (top) and $[4\text{Fe}-4\text{S}]^{4+}$ (bottom) states.

To describe the deviation from a reference spin density pattern, we introduce a spin density metric (SDM) defined as the Frobenius norm of the deviation of the outer product (σ) of the UHF Mulliken spin densities from that of the ideal (denoted I in the equation below) spin densities shown in the middle panels of Figures 1 and 2:

$$\text{SDM} = \left\| \frac{(\rho \otimes \rho) - (\rho_I \otimes \rho_I)}{\rho_I \otimes \rho_I} \right\|_F = \left\| \frac{\sigma - \sigma_I}{\sigma_I} \right\|_F \quad (9)$$

UCCSD and UCCSD(T) calculations were performed with PySCF.⁸⁶ While initial ph-AFQMC studies of the He_2^+ cation (specifically the calculations presented in Figure 10) used the ipie program,^{52,91} the rest used an in-house code. ph-AFQMC calculations used an imaginary-time step of $\Delta\tau = 0.005$ and a Cholesky decomposition threshold of 10^{-12} . We initialized the walkers with the same UHF state used as trial wave function throughout. For $[2\text{Fe}-2\text{S}]^{2+}$ and $[4\text{Fe}-4\text{S}]^{2+}$ autocorrelation effects were removed by a reblocking procedure where the optimal equilibration time was chosen such that the lowest ph-AFQMC statistical error is obtained.

To quantitatively assess the stochastically projected many-body wave function, $|\Psi_{\text{MC}}(\tau)\rangle$, we have computed the normalized overlap squared,

$|\langle \bar{\Psi}_T | \bar{\Psi}_{\text{MC}}(\tau) \rangle|^2$, with the trial wave function (or another wave function of interest) using the following equation:

$$|\langle \bar{\Psi}_T | \bar{\Psi}_{\text{MC}}(\tau) \rangle|^2 = \frac{|\langle \bar{\Psi}_T | \Psi_{\text{MC}}(\tau) \rangle|^2}{\langle \bar{\Psi}_T | \bar{\Psi}_T \rangle \langle \Psi_{\text{MC}}(\tau) | \Psi_{\text{MC}}(\tau) \rangle} \quad (10)$$

Our implementation exports all of the walkers $|\varphi_i(\tau)\rangle$ and associated weights $w_i(\tau)$ every 0.2 Ha^{-1} . At each of these imaginary-time slices, we evaluate the numerator in eq 10 as

$$|\langle \bar{\Psi}_T | \bar{\Psi}_{\text{MC}}(\tau) \rangle|^2 = \left| \sum_i c_i(\tau) \langle \bar{\Psi}_T | \varphi_i(\tau) \rangle \right|^2 \quad (11)$$

having defined the expansion coefficients $c_i(\tau) = w_i(\tau) / \langle \bar{\Psi}_T | \varphi_i(\tau) \rangle$, and the AFQMC wave function norm in the denominator as

$$\langle \Psi_{\text{MC}}(\tau) | \bar{\Psi}_{\text{MC}}(\tau) \rangle = \sum_i \sum_j c_i^*(\tau) c_j(\tau) \langle \varphi_i(\tau) | \varphi_j(\tau) \rangle \quad (12)$$

For these overlap calculations, populations of 1024 and 2048 walkers were used for He_2^+ and the Fe-S clusters, respectively. Statistical error bars were estimated by running multiple repeats (each with a different random number seed) and averaging the squared normalized overlaps

at each τ value across the repeats, yielding mean values and standard errors.

3. RESULTS AND DISCUSSION

3.1. $[2\text{Fe}-2\text{S}]^{2+}$

We begin by investigating the low-lying UHF states of the diferric complex, $[2\text{Fe(III)}-2\text{S}]^{2+}$. Figure 3 shows the

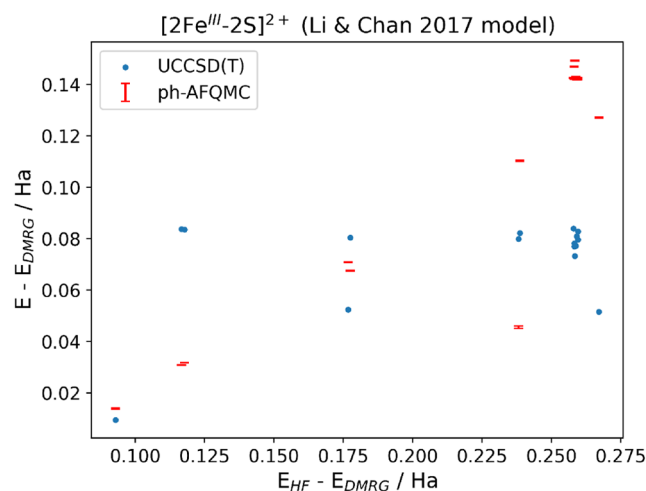


Figure 3. Deviations of the ph-AFQMC-UHF and UCCSD(T) energies from the DMRG reference, as a function of the deviation of the UHF trial/reference energy from DMRG.

deviations of the resulting ph-AFQMC and UCCSD(T) energies from the DMRG reference value, plotted against the energy error of the UHF trial/reference. The ph-AFQMC energy when using the lowest-energy UHF state deviates from the reference value by 13.9 ± 0.2 mHa; this result from a single determinant trial wave function is encouraging and arguably unexpected, given previous studies that required on the order of a million determinants to achieve a similar level of accuracy.⁵⁰ UCCSD(T) is slightly more accurate with this UHF reference state, above DMRG by 9.6 mHa, while the UCCSD energy

deteriorates to a 25.1 mHa error. Figure 3 shows that UCCSD(T) is relatively insensitive to the reference used, with most UCCSD(T) energies clustering between 52 and 84 mHa above DMRG despite more significant differences in the underlying UHF energies. That the UCCSD(T) energy using the lowest-energy HF reference is singularly outside of this energy range, and is substantially improved with respect to the DMRG reference value, goes against the common wisdom that CC is “orbitally insensitive.”⁹²

The experimentally deduced oxidation states and spin coupling pattern, along with the corresponding reference SDM, of the $[2\text{Fe}-2\text{S}]^{2+}$ cluster are shown in Figure 1. Figure S1 confirms the trend that lower energy UHF states correspond to more appropriate iron spin densities vs the idealized distribution. In this specific case the lowest-energy UHF trial wave function is the state with the smallest SDM value, which leads to the most accurate ph-AFQMC energy.

3.2. $[4\text{Fe}-4\text{S}]^{2+}$

We now consider the mixed-valence $[4\text{Fe}-4\text{S}]^{2+}$ cubane system, which was recently shown to have three low-lying spin isomers separated by at most a few mHa.³² Figure 4 shows the ph-AFQMC and UCCSD(T) energies corresponding to the different UHF states found, along with the theoretical best estimates for the three spin isomers. The UCCSD(T) energies form roughly a horizontal band ranging from 20 to 100 mHa above the references. In contrast, a nearly linear behavior can be seen in the case of the ph-AFQMC energies, as was observed also for $[2\text{Fe}-2\text{S}]^{2+}$ above. Notably, there is one UHF state that leads to remarkable agreement with the theoretical best estimates, which is *not* the lowest-energy UHF solution. While this result will be analyzed further in Section 3.4, we now look closer at how this optimal UHF trial might be identified among the myriad possible UHF states.

Focusing on what we will denote the $[+ - + -]$ spin isomer (which corresponds to the spin configuration shown in Figure 2), Figure 5 plots the ph-AFQMC and UCCSD(T) energy errors against the SDM of the UHF trial/reference wave functions, using the $[+ - + -]$ spin-density pattern as “ideal”. In the SDM range of 1–4.5 the trend in the ph-AFQMC energy

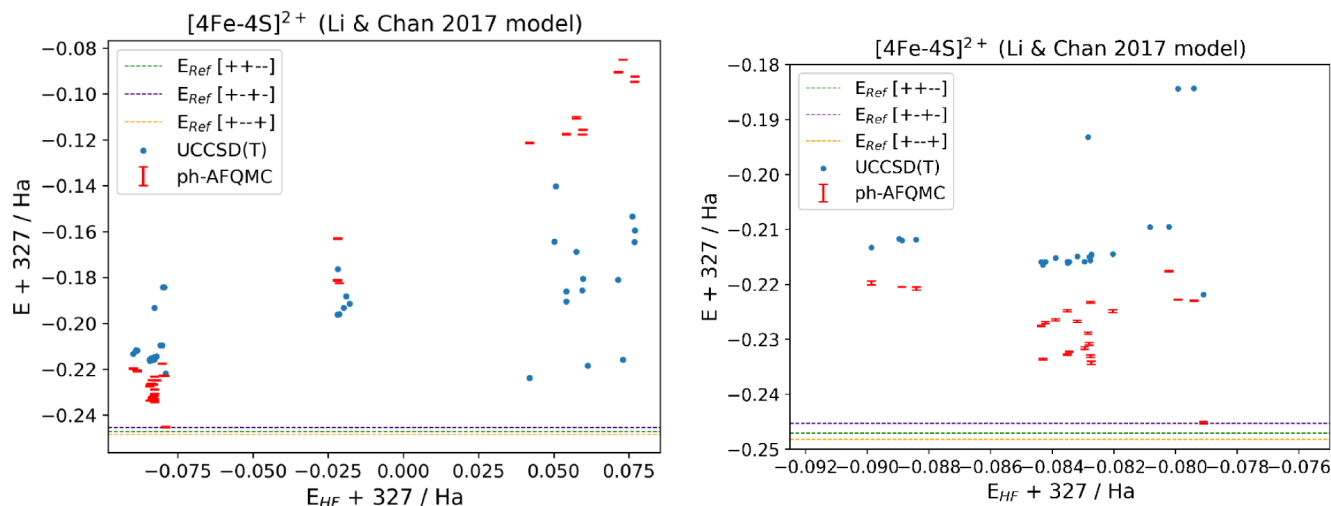


Figure 4. ph-AFQMC-UHF and UCCSD(T) energies as a function of the UHF trial/reference energy. Both panels show the same data, but the right panel is zoomed into the dense bottom left region. The three reference theoretical best estimates are $E_{\text{ref}}[+-+-] = -327.247062 \pm 0.000984$, $E_{\text{ref}}[+--+]$ = -327.245289 ± 0.000733 , $E_{\text{ref}}[+--+]$ = -327.248199 ± 0.001820 reproduced from Zhai et al.,³² wherein the 3 solutions are referred to as BS1, BS2, and BS3 respectively (the error bars are omitted for ease of readability).

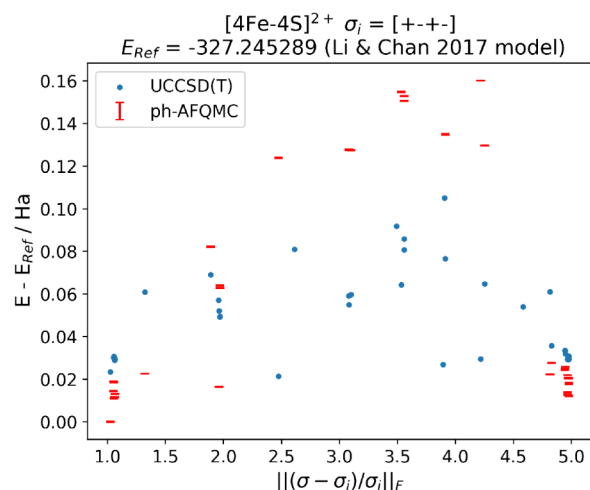


Figure 5. Deviations of the ph-AFQMC-UHF and UCCSD(T) energies from the theoretical best estimate³² of the $[+ - + -]$ spin isomer, as a function of the SDM of the underlying UHF state. We consider the ideal spin distribution, σ_i , to be $\text{Fe}_1^{12.5} \text{Fe}_2^{12.5} \text{Fe}_3^{12.5} \text{Fe}_4^{12.5}$.

error grows as the SDM increases. Moreover, it is clear that the UHF state with the lowest SDM is the best trial wave function, yielding a ph-AFQMC energy that agrees closely with the theoretical best estimate (and a well-behaved energy trajectory in imaginary-time). This optimal-SDM UHF trial is the 23rd excited UHF state among the UHF solutions that we found. Its atomic spin-densities are $+3.9, -3.9, +3.9, -3.9$ (where indices 1 and 3 correspond to the top face of the elongated cube, and indices 2 and 4 correspond to the bottom face). In a tetrahedral ligand field, Fe(II) and Fe(III) ions would be expected to correspond to 4 and 5 unpaired electrons, respectively; therefore the experimentally established 2.5 oxidation state would correspond to atomic spin density magnitudes of 4.5. Although the atomic spin density magnitudes are slightly smaller than 4.5, the relative sign structure matches the targeted $[+ - + -]$ spin isomer and the magnitudes on all four Fe ions are equal.

There is a cluster of UHF solutions that leads to competitively low ph-AFQMC energies in the SDM range 4.5–5.0. These do not show ferromagnetically coupled pairs in the top and bottom cube faces antiferromagnetically coupled, but rather other spin coupling patterns involving the four Fe ions (as opposed to charge excitations from, e.g., the bridging sulfur atoms). These are representative of other Hamiltonian eigenstates that are nearly degenerate with the ground-state, i.e., precisely the other two spin isomers that we will denote $[+ + - -]$ and $[+ - - +]$. To target these states, we constructed ideal atomic spin densities corresponding to these two spin isomers, and select the UHF trial/reference with the lowest SDM. Figure 6 shows the resulting ph-AFQMC and UCCSD(T) energies along with the theoretical best estimates for all three spin isomers. We find that ph-AFQMC with the optimal-SDM UHF trials in the cases of $[+ + - -]$ and $[+ - - +]$ leads to energies that are more positive than the reference values by 14.0 ± 0.2 and 28.5 ± 0.4 mHa, respectively. Taken together, these results suggest that the near-perfect accuracy found for the $[+ - + -]$ spin isomer cannot be generally expected for all Fe–S systems using ph-AFQMC with optimal-SDM UHF trials. Nevertheless, this ph-AFQMC protocol for the three $[\text{4Fe-4S}]^{2+}$ spin isomers clearly outperforms UCCSD(T).

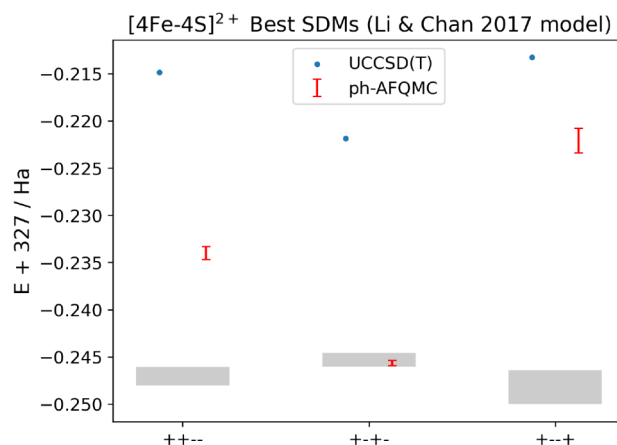


Figure 6. ph-AFQMC and UCCSD(T) energies for the three spin isomers, using the UHF trial/reference state with the lowest spin-density metric. The gray boxes indicate the theoretical best estimates from ref 32: the top of the box represents the upper-bound derived from a UCC procedure, the bottom represents the lowest of the extrapolated UCC or UDMRG procedures from the same work. Note that the SDM-optimal UHF employed for the $[+ - - +]$ spin isomer corresponds to the lowest-energy UHF solution identified for this system.

3.3. $[\text{4Fe-4S}]^{4+}$

Turning now to the tetraferrocene $[\text{4Fe(III)-4S}]^{4+}$ state, Figure 7 shows that the nearly linear trend of increasing ph-AFQMC

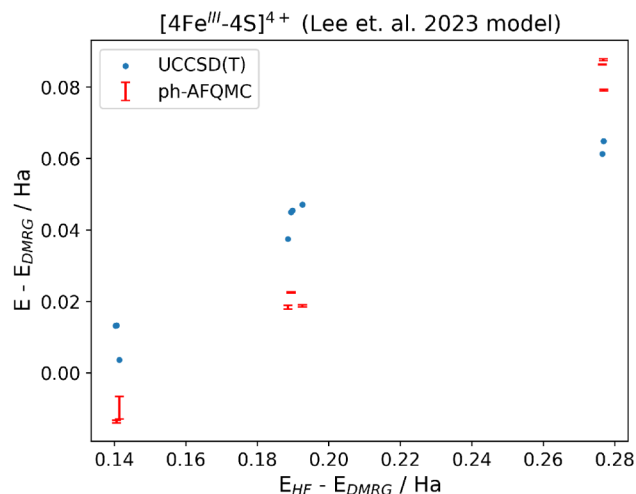


Figure 7. Deviations of the ph-AFQMC-UHF and UCCSD(T) energies from the DMRG reference, as a function of the deviation of the UHF trial/reference energy from DMRG.

energy error vs trial energy error is present here as well. Notably, there is a UHF reference that allows UCCSD(T) to be very accurate, deviating from the DMRG reference by only 3.6 mHa. The two lowest-energy UHF states, when used as trial wave functions in ph-AFQMC, result in energies that are too negative compared with the DMRG reference value.

There are three possible reference SDMs derived from the $[+ + - -]$, $[+ - + -]$, and $[+ - - +]$ orientations. Figure 8 takes the second as the reference SDM (choosing the other two leads to similar conclusions). The two UHF trials with extreme values of SDM, i.e., around 1 and 5, lead to the most accurate ph-AFQMC energies: 9.7 ± 3.1 and 13.6 ± 0.4 mHa below the DMRG energy, respectively. Furthermore, the unusual energy vs

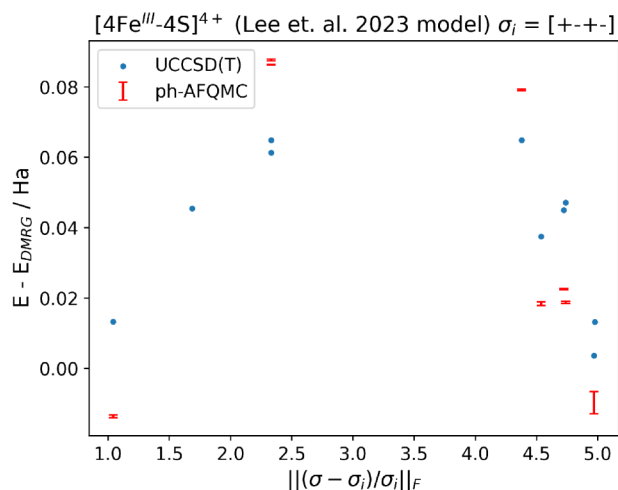


Figure 8. Deviations of the ph-AFQMC-UHF and UCCSD(T) energies from the DMRG reference, as a function of the SDM of the underlying UHF state. In this system we consider the ideal spin distribution, σ_i , to be $\text{Fe}_1^{\uparrow 3} \text{Fe}_2^{\downarrow 3} \text{Fe}_3^{\uparrow 3} \text{Fe}_4^{\downarrow 3}$.

imaginary-time trajectory when using the UHF trial with SDM ~ 1 is shown in Figure 9. The trajectory very quickly arrives at the

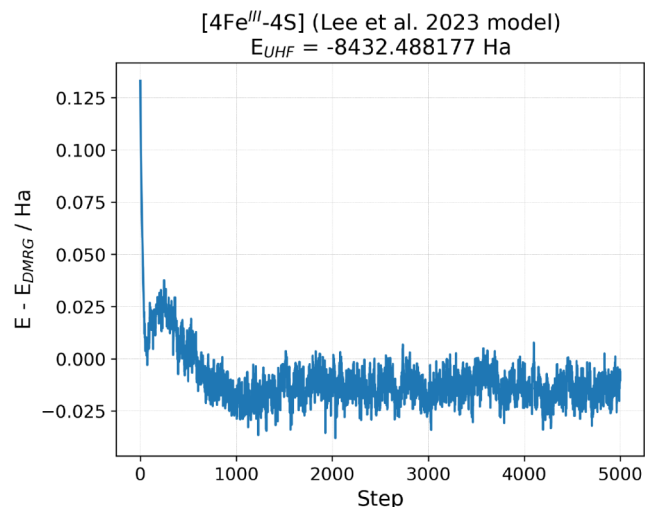


Figure 9. ph-AFQMC trajectory using the UHF trial with smallest SDM for the $[\text{4Fe-4S}]^{4+}$ state. Note the shallow dip early in the trajectory with a minimum near the reference DMRG value before a sharp rise and then slow decay to a relatively noisy plateau at an overly negative energy.

exact energy before collapsing to a final plateau that is too low in energy. While ph-AFQMC is not variational, we will show later that an unexpected mechanism involving small local energy denominators is responsible for the overly negative ph-AFQMC energies (and the unexpected features in the energy trajectory in this case).

The UHF state with the smallest SDM has atomic spin densities on the irons of magnitude 4.3. The Fe(III) oxidation state implies 5 unpaired same-spin electrons per Fe ion, so here again the UHF spin density magnitudes are slightly smaller. The electron configuration represented by this trial state is only one of six possible (collinear) configurations that place (approximately) five aligned local spins on each of the four Fe centers while respecting the constraint of $S_z = 0$. Unlike the coupling

patterns possible in the mixed-valence case, in which a ferromagnetic pair of Fe atoms is antiferromagnetically coupled to the other pair, each ferric center of the $[\text{4Fe(III)-4S}]^{4+}$ complex interacts through a “superexchange” mechanism^{93–96} to each of the others favoring antiferromagnetic coupling, resulting in geometric spin frustration wherein none of the local spins can be antialigned with all of its neighbors. Despite the slight structural distortion of the cubane, it is well-established that such frustration gives rise to near-degeneracies of the many-body states, with six spin configurations dominating the low-energy manifold. This implies that the overlap of the trial with the exact ground-state eigenfunction will be at most one-sixth (at most because other charge-excited/transferred configurations that contribute to the superexchange mechanism are not accounted for, and the UHF trial is strongly spin contaminated while the eigenstates are not).

3.4. Analogy with Stretched He_2^+

The dissociation of the helium dimer cation, He_2^+ , is a challenging model system for single-configurational electronic structure methods. The lowest-energy UHF solution exhibits spontaneous symmetry breaking, artificially localizing the single unpaired electron onto one of the Helium atoms. This UHF state has a dipole moment whereas the true wave function, which involves a linear combination of both possible localized solutions, does not. With approximate density functionals such as PBE and PBE0, self-interaction errors lead to a single-configurational state with zero dipole moment due to an unphysically delocalized electron/hole.^{97,98}

In this study we have identified two low-lying UHF solutions which energetically separate at inter-Helium distance $R > 1.75 \text{ \AA}$. The lowest energy state (denoted here as UHF_S), stable to orbital rotations, has a growing dipole moment along the dissociation coordinate whereas the unstable higher-energy state (UHF_U) retains a dipole moment of exactly zero.

Figure 10 shows the resulting ph-AFQMC dissociation curves, along with results using minimal (3e2o) CASSCF trials and also single determinant trials formed from occupied Kohn–Sham orbitals obtained with the PBE and PBE0 functionals. While UHF_U leads to under-correlated energies, which eventually plateau at $\sim 10 \text{ mHa}$ above the exact value by 3 $\text{\AA}$, UHF_S (the trial state with the growing dipole) leads ph-AFQMC to overcorrelate – at worst by some 30 mHa just before 4 $\text{\AA}$, before returning toward the ph-AFQMC energies obtained from the UHF_U trial. The energy vs imaginary-time trajectory at a bond length of 4.62 $\text{\AA}$ is shown in Figure 11; the ph-AFQMC energy quickly equilibrates to a transient plateau at the exact (FCI) energy, but then collapses to another lower-energy plateau that persists out to very long imaginary-times. This is exactly the same behavior as what was found previously for the $[\text{4Fe-4S}]^{4+}$ case. Note that the energies shown in Figure 10 were obtained by averaging only the second energy plateau (if two were encountered).

In the spirit of understanding the unexpected imaginary-time trajectory that results from use of the UHF_S state as the trial wave function, Figure 12 shows the energy gap between the two lowest-energy exact eigenstates, Ψ_0 and Ψ_1 , and their overlaps with the stable UHF_S trial wave function, as a function of bond distance. At very short distance, Ψ_0 and Ψ_1 are nearly degenerate but then separate to a maximum of roughly 0.754 Ha at 0.76 $\text{\AA}$. As the bond distance continues to increase, the energy gap decreases rapidly to again approach zero. Before 2 $\text{\AA}$, the squared overlap of the UHF_S trial with the exact ground-state,

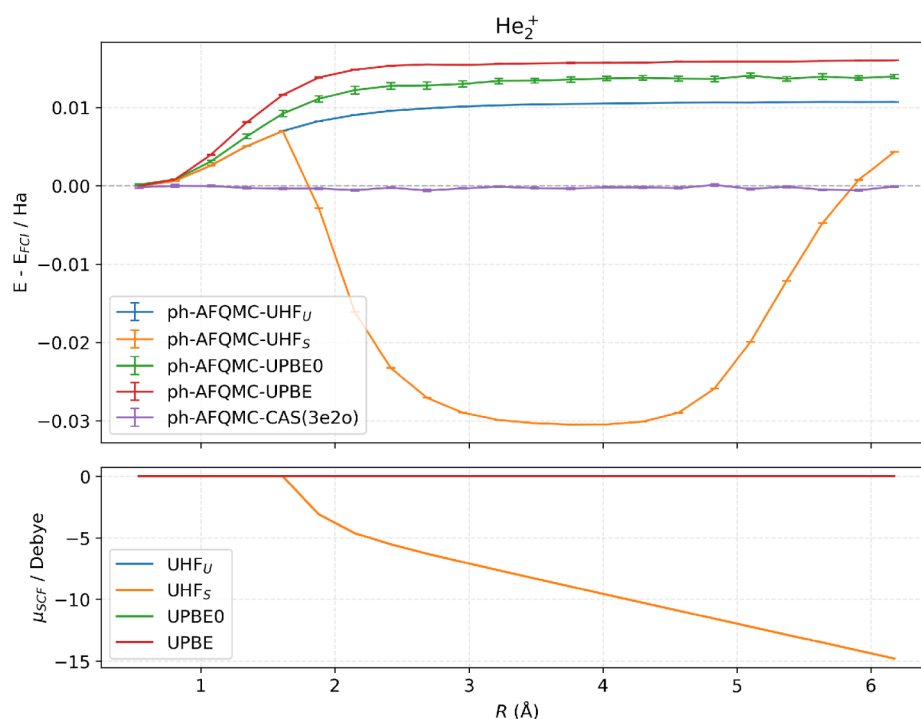


Figure 10. ph-AFQMC energies corresponding to various trial wave functions vs FCI along the dissociation coordinate of the helium dimer cation (top), and dipole moments of the single-determinant UHF and KS-DFT trials (bottom).

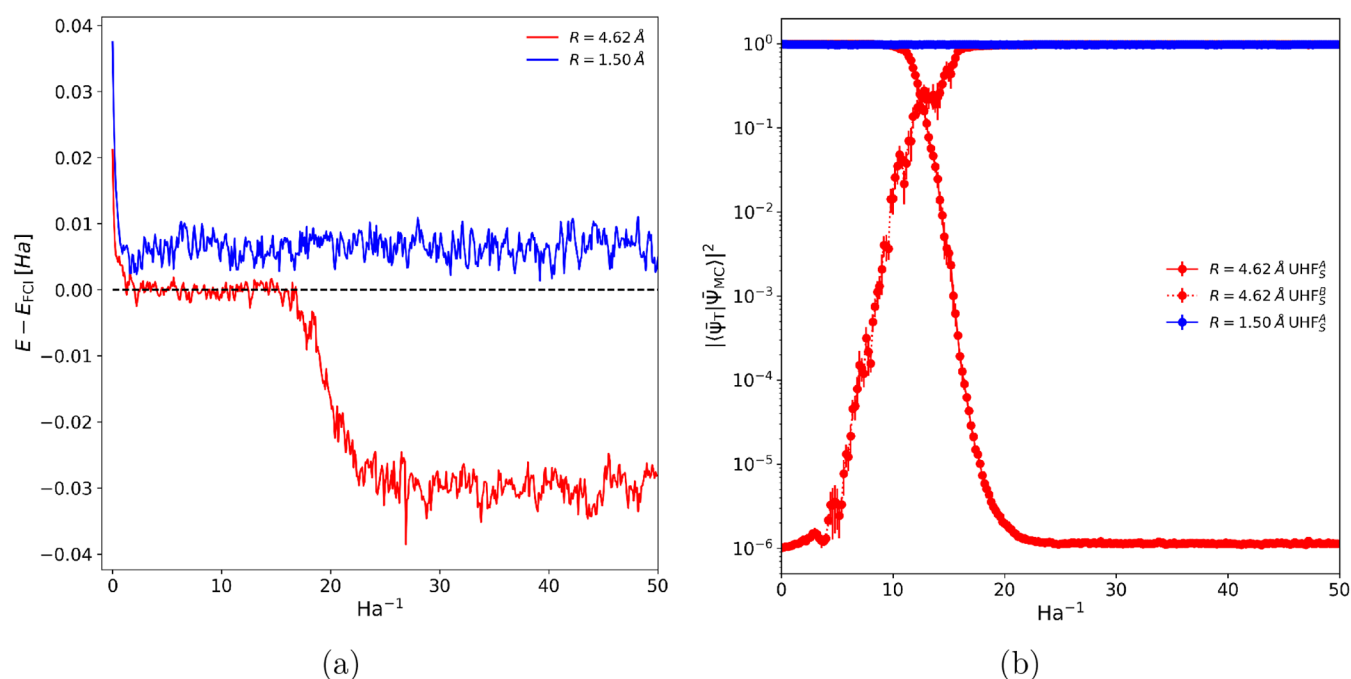


Figure 11. (a) ph-AFQMC energy trajectories for the He_2^+ system at $R = 1.50 \text{ \AA}$ and $R = 4.62 \text{ \AA}$ with the UHF_S trial wave function. Note the double-plateau structure for $R = 4.62 \text{ \AA}$. (b) Normalized squared overlaps between the stochastic ph-AFQMC wave function and UHF states at the two bond lengths. Mean values and uncertainties at individual points were obtained from averaging 4 independent runs. Note the logarithmic y axis.

$|\langle \Psi_{\text{UHF}_S} | \Psi_0 \rangle|^2$, is close to one; just before $2 \text{ \AA}$ this quantity starts to drop while $|\langle \Psi_{\text{UHF}_S} | \Psi_1 \rangle|^2$ increases correspondingly; by $4 \text{ \AA}$ both squared overlaps are roughly equal at 0.5. The abnormal ph-AFQMC trajectories, which are observed at roughly $2 \text{ \AA}$ and beyond, would seem to coincide with the onset of a nonzero squared overlap of the trial with the first excited doublet

eigenstate, and when Ψ_0 and Ψ_1 become nearly degenerate in energy.

For further analysis we must now acknowledge that UHF_S is doubly degenerate as one could localize the lone electron on either Helium atom (A or B), denoted as UHF_S^A and UHF_S^B. As the He atoms separate, the two FCI eigenstates are $\pm$ combinations of these two UHF states with the hole localized

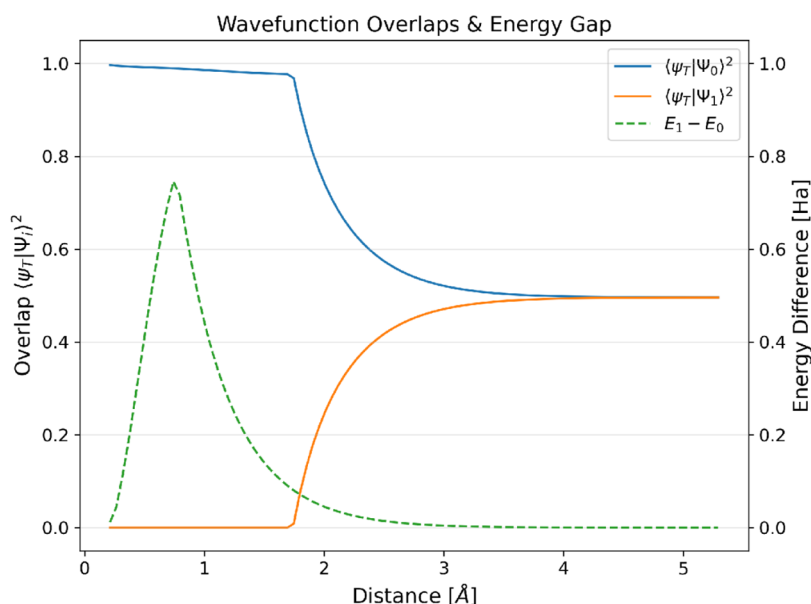


Figure 12. For the dissociation of He_2^+ , the energy gap between the lowest and first-excited doublet eigenstates (right vertical axis) and the squared overlap of the lowest-energy UHF trial wave function (UHF_S) with the two eigenstates (left vertical axis) are shown.

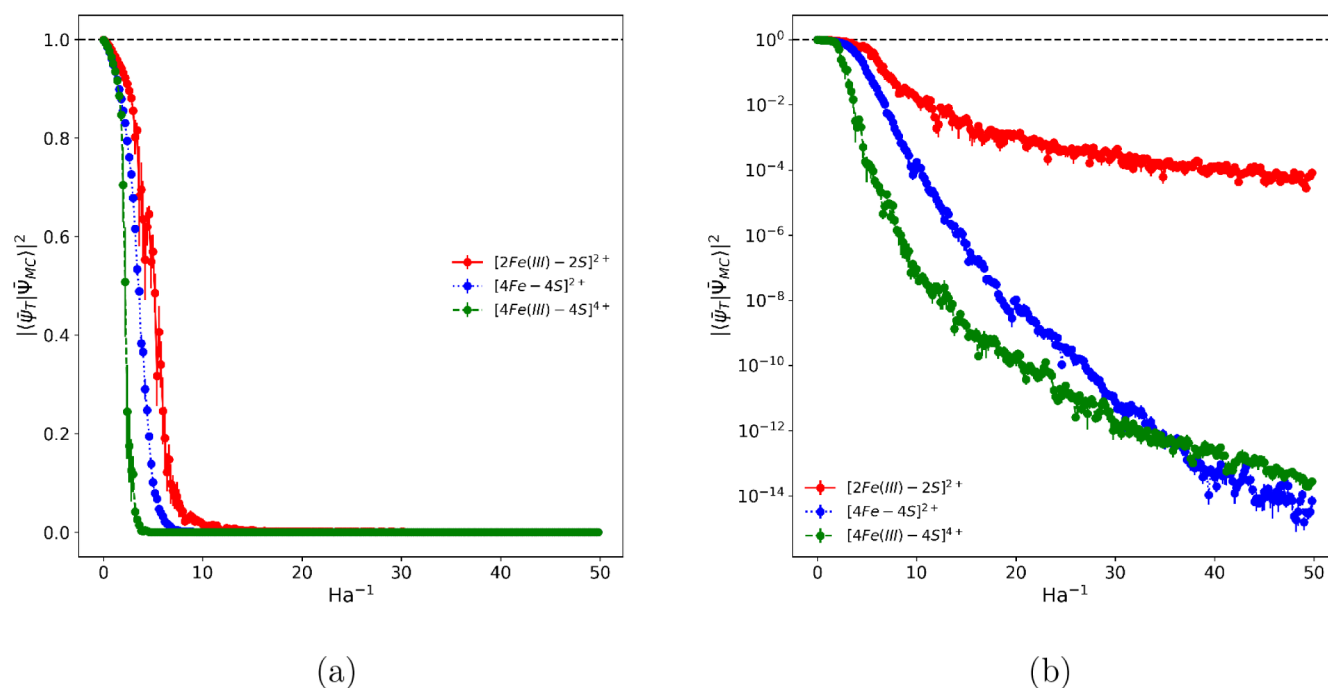


Figure 13. Normalized squared overlaps between optimal-SDM UHF trial and the respective ph-AFQMC wave function for each iron–sulfur cluster studied in this work. Mean values and uncertainties at individual points were obtained from averaging 4 independent runs. Linear y-axis (a) and logarithmic y-axis (b).

on different He atoms. It is therefore reasonable to expect that a trial that consists of only a single configuration and does not encode knowledge of the correct relative phase in the FCI ground eigenstate expansion could lead to suboptimal sampling in ph-AFQMC; specifically we propose that the overly negative ph-AFQMC energy in the intermediate region with the UHF_S trial must be due to overly negative walker local energies.

Consider a walker which starts out close to the UHF_S^A trial; as the ph-AFQMC propagation is simply a string of one-body operators, it will be very probable for one or more walker to

evolve to have the hole localized on atom B (UHF_S^B). The walker's local energy in this situation becomes

$$E_L = \frac{\langle \text{UHF}_S^A | \hat{H} | \text{UHF}_S^B \rangle}{\langle \text{UHF}_S^A | \text{UHF}_S^B \rangle} \quad (13)$$

At intermediate distances, the denominator will be small and the numerator will be negative, which would account for the overly negative local energies observed. Furthermore, since the weights are proportional to $e^{-\Delta\tau E_L}$, the population control algorithm – which periodically duplicates walkers with large

weights and removes walkers with small weights – will accelerate the growth of a large population of walkers with determinants in the UHF_S^B state. Indeed, this behavior is confirmed in panel (b) of Figure 11, where the overlap of the ph-AFQMC wave function with the UHF_S^A trial at $R = 4.62 \text{ \AA}$ collapses to around 10^{-6} while the overlap with UHF_S^B rises, coinciding with when the energy trajectory drops toward the second plateau. In contrast, for $R = 1.50 \text{ \AA}$, the ph-AFQMC wave function retains a large overlap with the UHF_S^A trial and the energy trajectory is nonpathological.

To verify whether or not similar walker dynamics are at play for the three iron–sulfur clusters, we computed the overlaps of the ph-AFQMC wave functions with the SDM-optimal UHF trials discussed above. As shown in Figure 13a, the overlaps decay to very small values in all cases early in the imaginary time propagation, well before the energy equilibrates. Closer examination of the overlap at longer imaginary time and on a logarithmic scale (Figure 13b) reveals a qualitative difference between $[\text{2Fe(III)}-2\text{S}]^{2+}$ and the other two clusters, with the former's overlap plateauing at roughly 10^{-4} instead of approaching a significantly smaller value of order 10^{-15} (consistent with the difference in system sizes). While these UHF trials led to reasonably accurate ph-AFQMC energies, they evidently do not effectively constrain the walkers to retain a high overlap with the trial, unlike the well-behaved example of He_2^+ at $1.50 \text{ \AA}$. The multitude of possible UHF states for the iron–sulfur clusters makes determining which of them, if any, might have a growing or large overlap with the AFQMC wave function challenging and will be the focus of future investigation.

Nevertheless, the decay of the overlap between the stochastically evolving wave function and the UHF trial suggests that, on average, small local energy denominators are causing overly negative local energies. This is a sampling bias that is distinct from the phaseless bias (indeed, they likely bias the energy in opposite directions). The dominance of this sampling bias especially in the cases of the $4\text{Fe}-4\text{S}$ cubanes would explain the overly negative ph-AFQMC energies found in our study of $[\text{4Fe}-4\text{S}]^{4+}$, and also previous literature reports of ph-AFQMC overcorrelating at stretched bond lengths (e.g., in N_2 , C_2 , and H_{10}) when UHF trials are used,^{45,99,100} and the convergence “from below” in the $[\text{2Fe}-2\text{S}]$ model from ref 50, with localized orbitals as the number of determinants in the trial was increased. Our overlap results imply that this sampling bias is at play even in $[\text{2Fe}-2\text{S}]^{2+}$ and $[\text{4Fe}-4\text{S}]^{2+}$ where the optimal-SDM UHF trials led to ph-AFQMC energies above or exactly on top of the theoretical best estimates, albeit possibly to a lesser extent given that the energy trajectories equilibrate monotonically and are relatively well-behaved.

4. CONCLUSIONS

In this work we have considered three iron–sulfur active space models derived from synthetic, biomimetic clusters – $[\text{2Fe}-2\text{S}]^{2+}$, $[\text{4Fe}-4\text{S}]^{2+}$, and $[\text{4Fe}-4\text{S}]^{4+}$ – that were designed to capture the valence ($\text{Fe } 3d$ and $\text{S } 3p$) electron correlation effects at play in many biological metalloenzymes. Accurately solving for the energies of the strongly correlated low-lying states represents a challenging frontier for approximate electronic structure theories. One manifestation of the challenges associated with these clusters is the existence of multiple mean-field solutions in close energetic proximity. Therefore, a key question for correlated methods like phaseless AFQMC or

Coupled Cluster approaches is which SCF solution to pick as a trial or reference state. Our work suggests that this selection ought to consider not just energies but also properties that can be deduced from physical symmetries, including, but not limited to, those from group theoretic arguments and/or experimental spectroscopic measurements.

We show that for the three iron–sulfur systems in this study there exists a UHF solution that can lead to reasonably accurate ph-AFQMC and UCCSD(T) ground-state energies vs the theoretical best estimates, and highlight the sensitivity of both methods to the UHF trial/reference state employed. The lowest-energy UHF state does not always lead to the most accurate ph-AFQMC and UCCSD(T) energies. When informed by physical symmetries and ideally structural or spectroscopic characterization of the metal oxidation states and relative spin orientations, we demonstrate that a function capturing the deviation from ideality of the atomic Mulliken spin densities – which we call the spin density metric (SDM) – can reliably identify the optimal UHF trial/reference state for subsequent ph-AFQMC and CCSD(T) calculations. We note that the SDM utilized here for UHF trials is not a unique choice, and alternative physically transparent metrics such as spin–spin correlation functions, local $\hat{S}_z$ expectation values, etc., could also be employed. Using the optimal-SDM UHF trial enables unexpectedly high accuracy in specific cases: for one spin isomer in $[\text{4Fe}-4\text{S}]^{2+}$ ph-AFQMC is essentially exact, and UCCSD(T) is off by 3.6 mHa for $[\text{4Fe}-4\text{S}]^{4+}$. However, across all of the Fe–S systems investigated in this work the performance of ph-AFQMC with optimal-SDM UHF trials is more modest, showing average errors of ~ 10 – 20 mHa from the theoretical best estimates. Nevertheless given the low $O(N^4)$ scaling of ph-AFQMC-UHF (per-sample) with system size, vs the $O(N^7)$ and $O(e^N)$ scaling of UCCSD(T) and methods such as selected CI and DMRG, respectively, this level of accuracy for such challenging strongly correlated systems is encouraging.

A closer look at the ph-AFQMC trajectories reveals an unexpected deficiency in the ability of even the optimal-SDM UHF trial to act as an effective importance function in the case of these three Fe–S systems. Specifically, the average overlap of the stochastic QMC wave function with the UHF trial was found to decay rapidly, as was found also for the stretched helium dimer cation. The small local energy denominators result in pathological reweighting that leads eventually to an equilibrated walker population with very small overlaps with the UHF trial wave function, which tends to push the ph-AFQMC energy estimate too negative. In the case of $[\text{4Fe}-4\text{S}]^{4+}$ (also stretched He_2^+) we find nonmonotonic energy trajectories in imaginary time resulting in an overly negative ph-AFQMC energy, whereas in $[\text{2Fe}-2\text{S}]^{2+}$ and $[\text{4Fe}-4\text{S}]^{4+}$ the trajectories are not unusual and the artificial overcorrelation due to this sampling bias appears to be compensated (sometimes exactly) by the phaseless bias.

These observations have fundamental implications for how we can understand the role of the trial wave function in ph-AFQMC, and moreover how one might design improved classes of trial wave functions (at present using polynomial-cost classical algorithms but in the future potentially exploiting quantum computation/simulation^{101–106}). One takeaway is that when the true ground-state involves multiple large-weighted configurations superposed with specific relative phases (which, as most clearly seen in the case of stretched He_2^+ , can distinguish one low-energy eigenstate from another), the determinants, their

phase relationships, and moreover the requisite physical symmetries ought to be encoded in the trial wave function not only to mitigate the phaseless bias but also to realize properly functioning importance sampling.

Taken together, this work suggests that despite being able to yield highly accurate ph-AFQMC energies in certain cases, UHF wave functions are *not* entirely satisfactory as trial states when targeting the strongly correlated low-lying states of these active space Fe–S models. Indeed, it remains an open question how to systematically improve the trial wave function *a priori* for strongly correlated states (especially of large system sizes). A related study by Kjønsstad et al.⁵⁸ has found for various Fe–S active space models that using CI subspace-projected coupled cluster expansions on top of UHF references does not consistently lead to systematic improvement in the ph-AFQMC energy even as higher order coupled cluster wave functions (of increasing fidelity with respect to the exact or near-exact eigenstate) are used. Our present work, which reveals a sampling bias when UHF trials are used that spuriously leads to overly negative ph-AFQMC energies, provides an explanation for why ph-AFQMC energies with subspace-projected UCCSD and (for the [4Fe–4S] case) UCCSDT trials are more positive than those obtained with ph-AFQMC-UHF. Our group is currently pursuing alternative routes to trial wave function improvement involving, e.g., self-consistency^{107,108} and the breaking and/or restoring of a hierarchy of symmetries.^{109,110} Future work should also explore different geometries¹¹¹ and charge states of these and other Fe–S clusters,⁶¹ and ideally go beyond active space models.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and the Supporting Information files, which together specify all parameters and protocols needed to independently reproduce the computed results. All UHF and CC calculations were performed using the open-source PySCF code. Regarding AFQMC, the data in Figure 10 were obtained using the open-source AFQMC code, ipie. Although the rest of the AFQMC calculations were performed using an in-house program, which cannot be made available at this time due to third-party interests, we have provided all the information needed for others to reproduce the published AFQMC energies with publicly available AFQMC codes. We also provide the script to statistically analyze the energy versus imaginary-time trajectories.

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jctc.6c00899>.

Plot of AFQMC energies versus the SDM for the [2Fe–2S]²⁺ cluster (PDF)

1-Electron reduced density matrices for all UHF solutions investigated for all Fe–S clusters; FCIDUMP integrals for the all-ferric 4Fe–4S system; representative AFQMC input and output file (corresponding to Figure 9); Python script used to reblock AFQMC outputs; spreadsheet file containing all of the raw energies along with UHF $\langle S^2 \rangle$ and Fe spin density values; FCIDUMP integral files for the 2Fe–2S and mixed-valence 4Fe–4S systems can be found on the following publicly accessible GitHub page: <https://github.com/zhendongli2008/Active-space-model-for-Iron-Sulfur-Clusters/tree/main> (ZIP)

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Notes

The authors declare no competing financial interest.

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